

Descriptive Analytics Report: PhonePe

Daily Transaction Behavior at Scale

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Tracking Transaction Trends, Payment Categories, Success Rates, Regional Demand & Operational Status

Note on data: This report uses the supplied PhonePe dataset containing 200 records across 5 dates, 8 states and 6 payment categories. The dataset is already aggregated rather than raw event-level transaction data. Therefore, the analysis describes the supplied observations and does not assume a larger real-time PhonePe transaction stream.

1. Objective

Build a descriptive analytics layer over PhonePe transaction data to answer following recurring business questions:

- How are daily transaction volumes trending over time?
 - How are daily GMV trending over time?
 - Which payment category contributes the highest transaction volume?
 - Which payment categories contribute the most to overall GMV?
 - Which states generate the highest transaction volumes?
 - Which states contribute the most GMV?
 - Successful transactions level.
 - State level transaction & Transaction demand geographically.
 - How does active merchant activity relate to transaction volume?
 - Weekday and Weekend transaction patterns.
 - Customer Satisfaction (CSAT).
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2. How This Would Work at Real Scale

Processing large-scale digital-payment events is an engineering problem before it is an analytics problem. A realistic architecture would collect transaction events, aggregate them by time/category/state, store raw and aggregated data, and expose the resulting KPIs through a Data visualization.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the transaction dataset	Google, Claude & Chat GPT
Stream processing	Aggregate by category, state and time	JUPYTER notebook (python)
Storage	Retain raw and processed data	Excel files/CSV files
Batch aggregation	Daily KPI and summary metrics	JUPYTER Notebook
Report	Present analysis, graphs and insights	Data Visualization/MS Word

3. Daily Transaction Volume

Over the supplied **5-day window**, the dataset records **23,853,873 transactions** and approximately **₹1,428.14 Cr in GMV**. The daily charts below show how transaction volume and GMV move across the available dates.

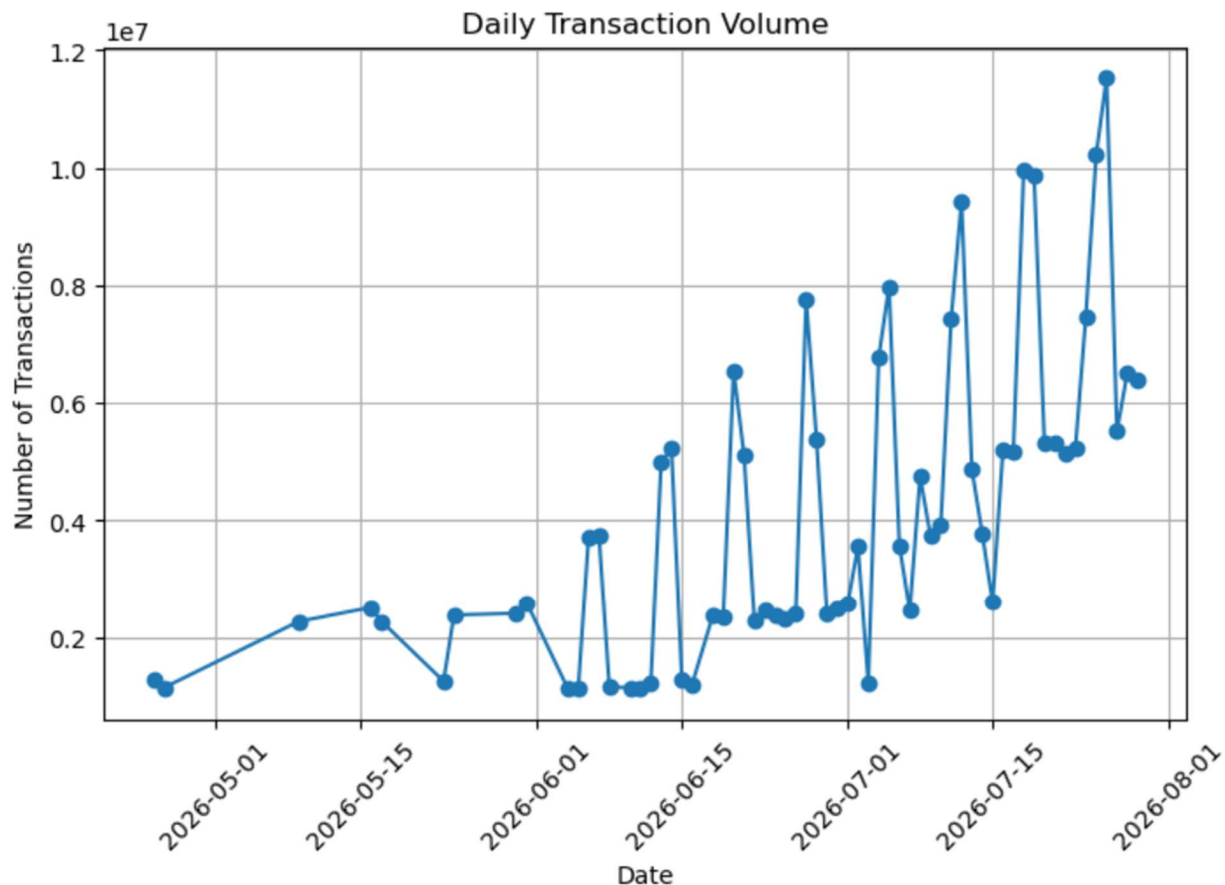


Figure 1. Daily transaction volume

INSIGHTS:

- Higher-volume dates indicate periods of stronger transaction activity.
- Lower-volume dates indicate comparatively weaker activity.
- Monitoring daily transaction volume can help identify demand fluctuations.
- This can support operational and capacity planning.

KPI :

Total transactions: **23.85 Million**

4. Daily GMV trending over time

Total GMV in the supplied records is **~105.40 Billion**.

The average GMV per represented date is **~1.673 Billion** while the highest daily aggregate GMV is **~4.614 Billion** GMV is calculated directly from the dataset's GMV field.

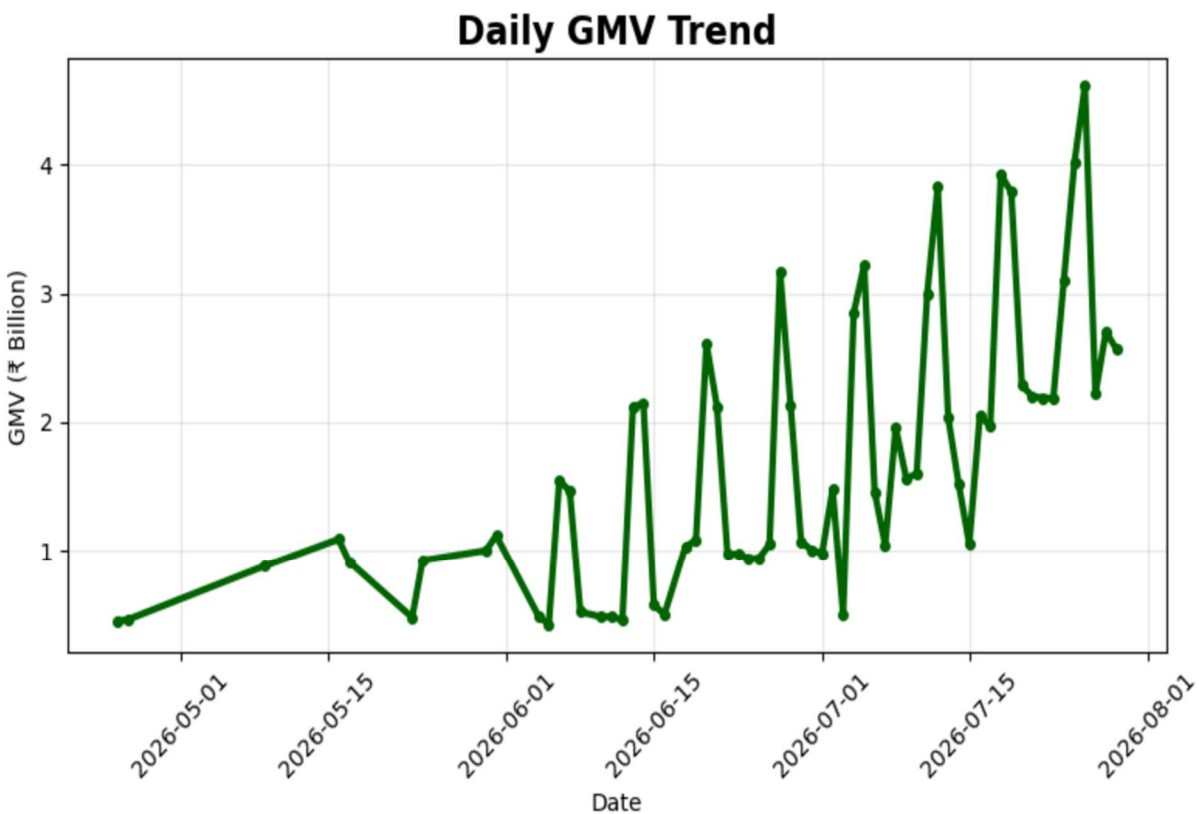


Figure 2. Daily GMV trend

INSIGHTS:

- GMV varies across the available dates.
- Higher GMV indicates stronger monetary transaction activity.
- Daily GMV monitoring helps identify high-value transaction periods.

- GMV should be considered alongside transaction volume to understand demand more completely.

KPI :

Total GMV: ₹ **105.40 Billion**

Average GMV per represented date: ≈ ₹**1.67 Billion**

Peak daily GMV: ≈ ₹**4.61 Billion**

5. Best-Performing Payment Categories

The uploaded Top **200** dataset contains only one payment category: **UPI - Merchant/QR**. It therefore accounts for **100%** of both transaction volume within this file. A comparison between multiple payment categories is not supported by the supplied rows and should not be inferred.

Transaction Share by Payment Category

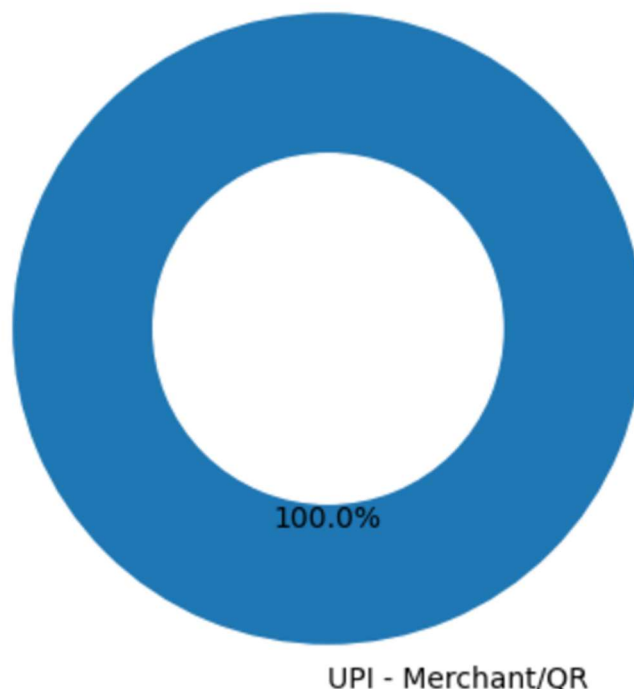


Figure 3. Transaction share by payment category

INSIGHTS:

- The graph does not provide a meaningful comparison between multiple payment categories.
- No conclusion should be made that UPI – Merchant/QR is the highest category across all PhonePe transactions.
- The result applies only to the supplied dataset.

KPI :

Top payment category: **UPI – Merchant/QR**

Transaction share: **≈100%**

Categories represented: **1**

6. Payment categories contribute the most to overall GMV

The uploaded Top **200 dataset** contains only one payment category: UPI - Merchant/QR. It therefore accounts for **100% of GMV** within this file. A comparison between multiple payment categories is not supported by the supplied rows and should not be inferred.

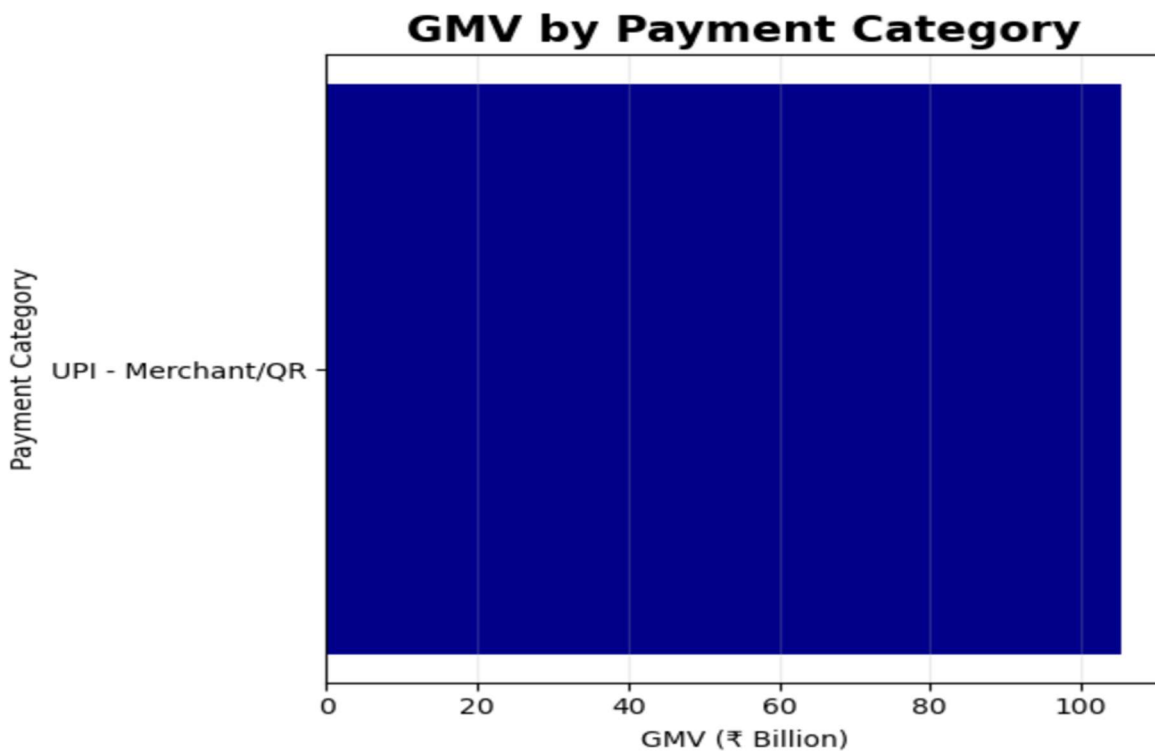


Figure 4. GMV by payment category

INSIGHTS:

- Therefore, the graph cannot establish which category contributes the most GMV among multiple categories.
- A broader dataset containing multiple payment categories would be required for that comparison.
- The finding should be interpreted only within the supplied dataset.

KPI :

Top GMV category: **UPI – Merchant/QR**

GMV share: **≈100%**

Categories represented: **1**

7. States generate the highest transaction volumes

Maharashtra is the largest state-level contributor by transaction volume, with **82,143,279 transactions (31.71% of all supplied transactions)**.

Maharashtra is also the largest contributor by GMV, at **33,588,996,843.15**.

The distribution shows that demand is concentrated in a subset of states rather than evenly distributed.

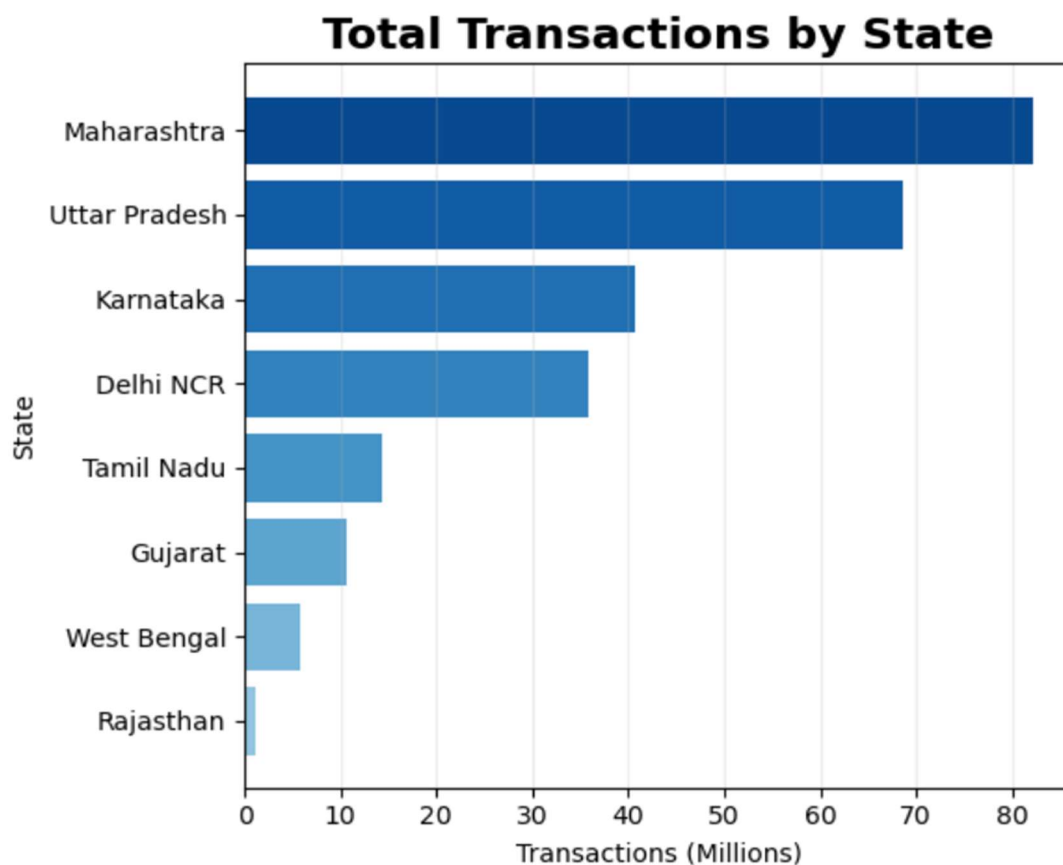


Figure 5. Total transaction by state

INSIGHTS:

- Transaction demand is therefore concentrated in a subset of states.
- Demand is not evenly distributed across all states.
- High-volume states may require greater operational and merchant support.

- State-level transaction analysis can help with regional business planning.

KPI :

Top-performing state: **Maharashtra**

Recorded transactions: **82.14 Million**

Share of total transactions: **≈31.71%**

8. States contribute the most GMV

Maharashtra is also the largest contributor by GMV, at ~ **33.59 billion**.

The distribution shows that demand is concentrated in a subset of states rather than evenly distributed.

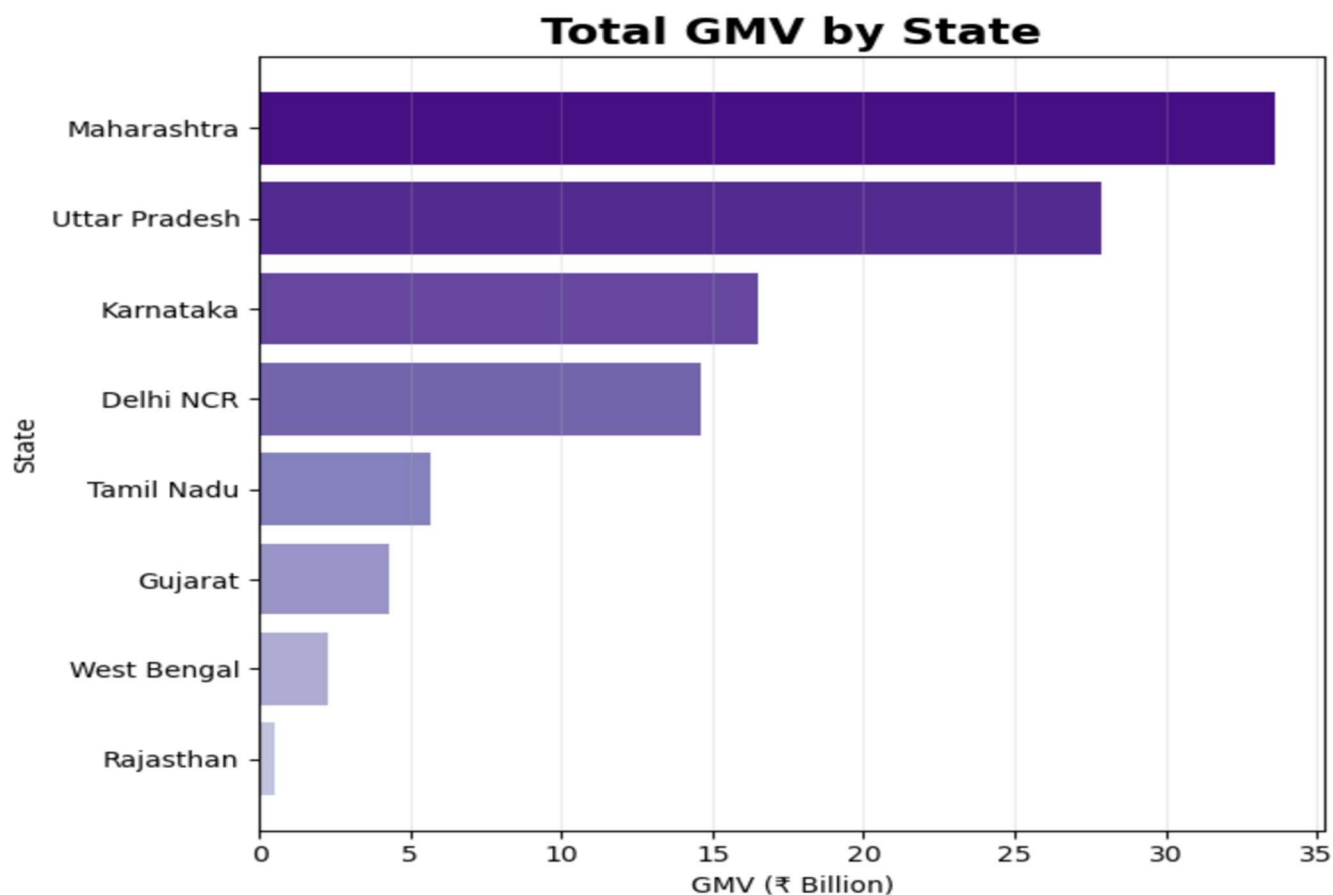


Figure 6. Total GMV by state

INSIGHTS:

- This indicates strong monetary transaction activity in the state.
- GMV is concentrated in a subset of states.
- High-GMV states can be important markets for operational and service planning.
- State-level GMV analysis can help identify commercially important regions.

KPI:

Top-performing state: **Maharashtra**

Recorded GMV: **≈₹33.59 Billion**

9. State level transaction demand

Maharashtra is the largest state-level contributor by transaction volume, with **82.14** transactions (**31.71% of all supplied transactions**).

Maharashtra is also the largest contributor by GMV, at **33.59 Billion**. The distribution shows that demand is concentrated in a subset of states rather than evenly distributed.

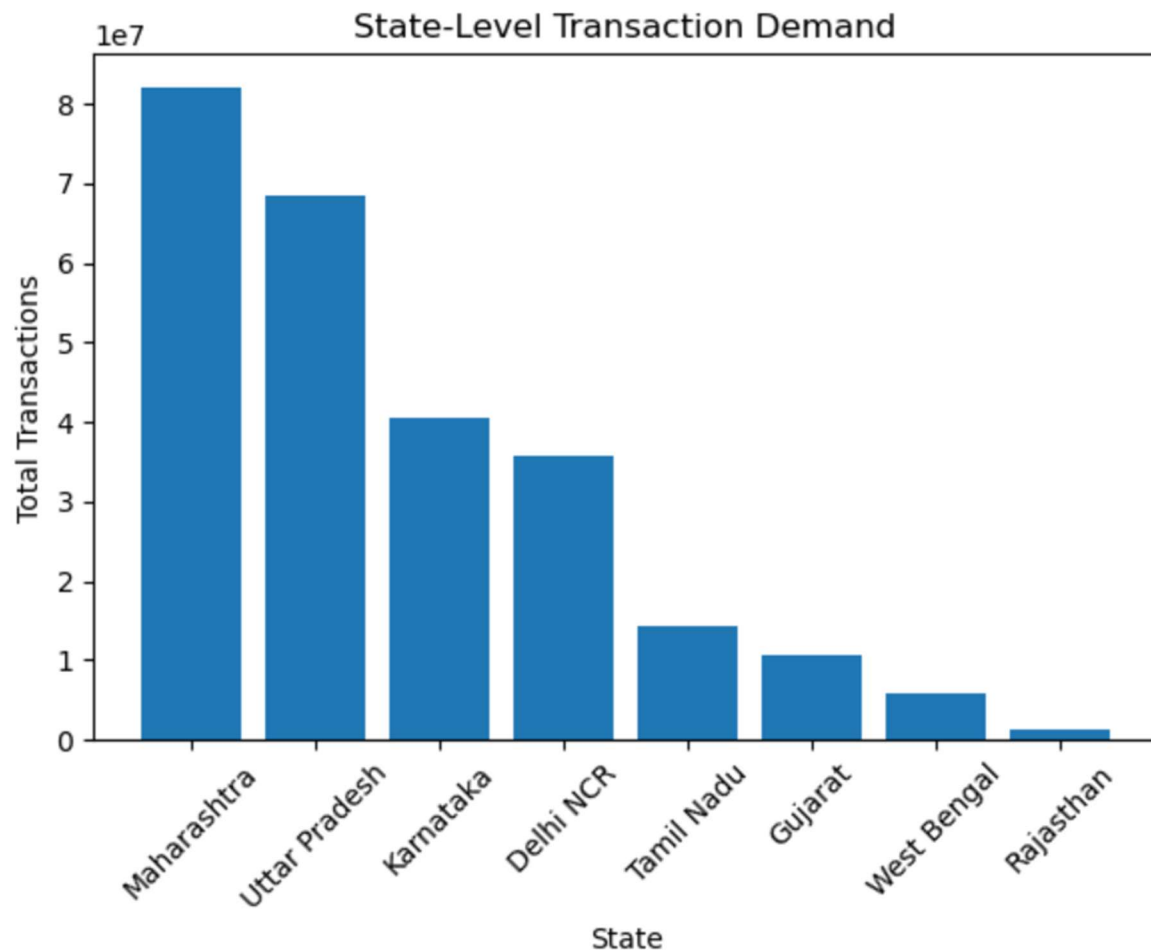


Figure 7. State Level Transaction Demand

INSIGHTS:

- Demand is distributed across **8 states**.
- Maharashtra therefore represents the strongest demand market in the graph.
- State-level demand information can support merchant coverage decisions.
- It can also support operational planning and service monitoring.
- Comparing states helps identify areas requiring greater business attention.

KPI:

Top-performing state: **Maharashtra**

Transaction volume: **3.80 Million**

States covered: **8**

10. Success Rate

The average transaction success rate across the supplied records is **97.96%**

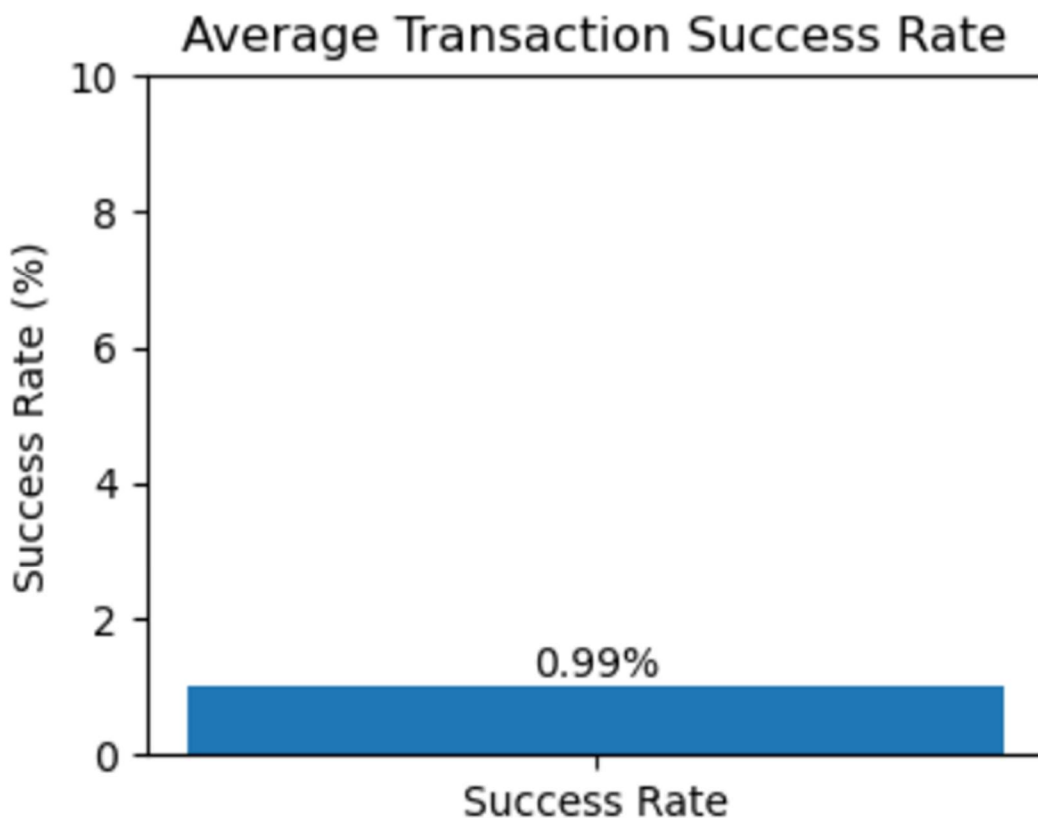


Figure 8. Average success rate by payment category

INSIGHTS:

- **State-level differences:** The state-wise graph can reveal which states have relatively higher or lower success rates. States below the overall average may need closer investigation.
- **Success rate** should be monitored continuously because changes may indicate operational issues.

- The average **CSAT score is 4.62/5**. Success rate is an operational reliability metric, whereas CSAT reflects the customer experience captured in the dataset.

KPI:

Average transaction success rate: **97.96%**

Operational status: **High transaction reliability**

Payment category: **UPI – Merchant/QR**

11. Regional Demand

Transaction demand is distributed across **8 states**. Maharashtra has the highest transaction volume in the supplied data at **3,802,979 transactions**. Regional demand analysis can support merchant coverage, operational planning, service monitoring and state-level business decisions.

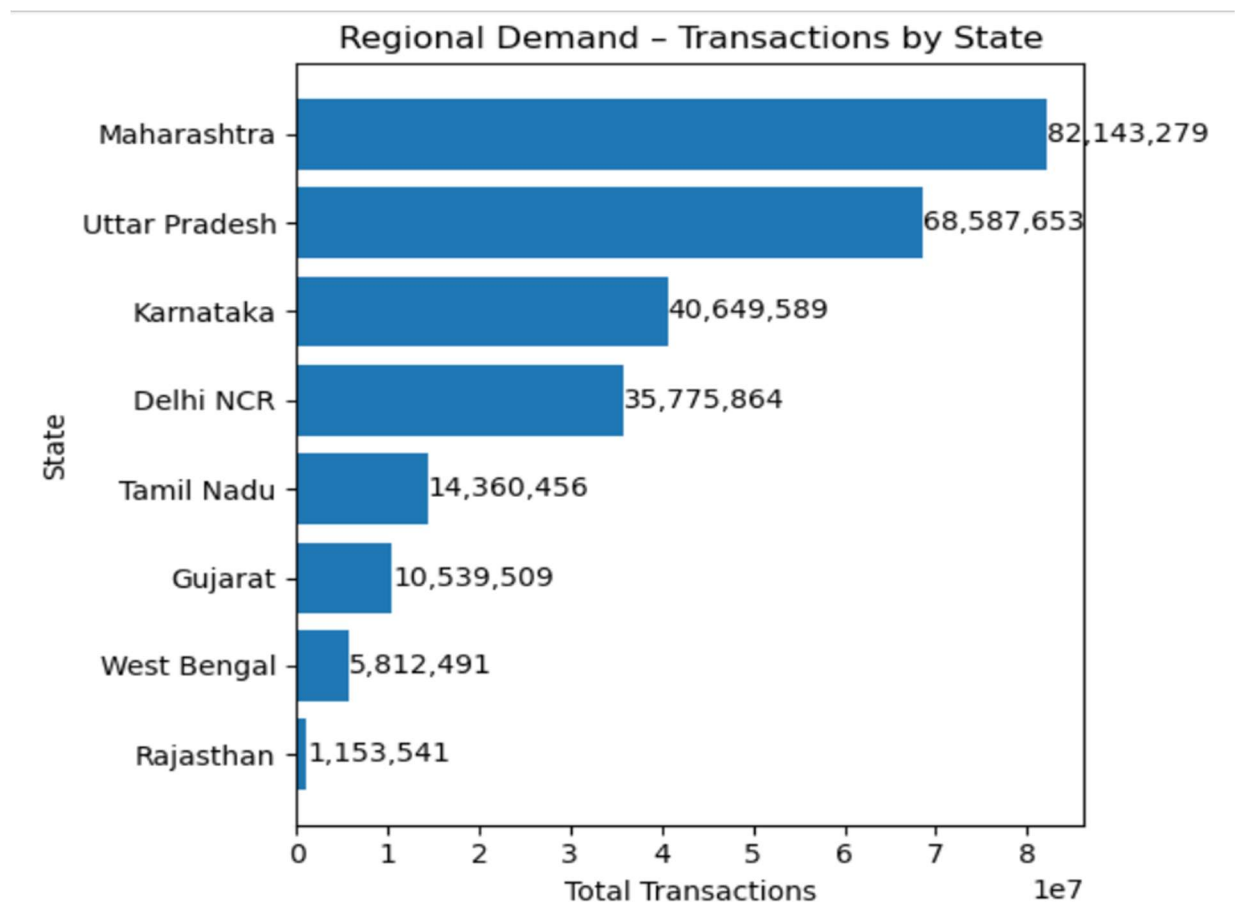


Figure 9. Total transactions by state

INSIGHTS:

- Regional demand can support merchant-coverage planning.
- It can also help with operational planning and service monitoring.
- State-level demand information can support regional business decisions.

12. Active Merchants VS Transaction Volume

The relationship between active merchants and transaction volume provides an additional descriptive view of state-level activity. States with more active merchants may generate higher transaction volumes, but the scatter plot should be interpreted as association rather than proof of causation.

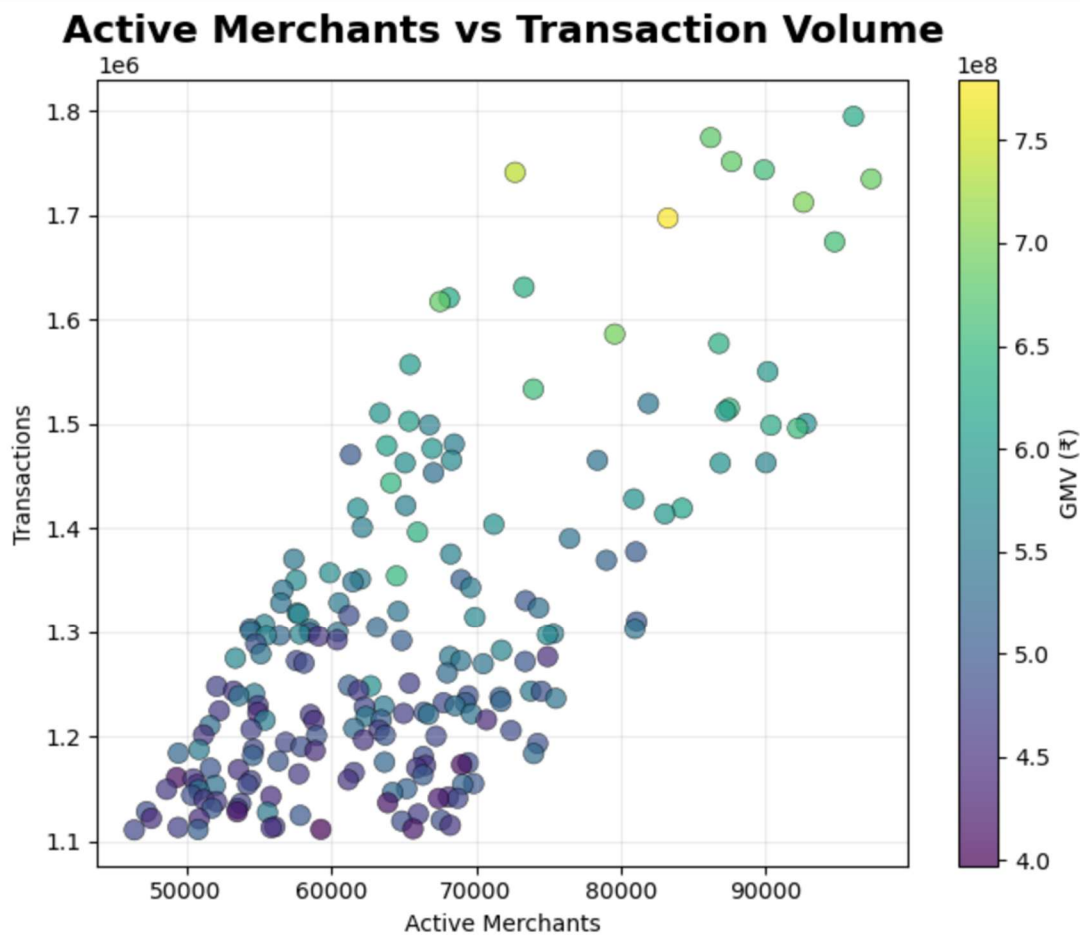


Figure 10. Active merchants vs transaction volume by state

INSIGHTS:

- A positive association may indicate that merchant coverage is related to transaction activity.
- However, the graph does not prove that more merchants cause higher transaction volume.
- Other factors can also influence transaction demand.
- This analysis can support merchant productivity and coverage planning.

KPI:

Top transaction state: **Maharashtra**

Active merchants in Maharashtra: **≈4.15 Million**

Transaction volume: **≈82.14 Million**

Relationship: **Higher merchant activity is associated with higher transaction volume in the supplied data**

13. Weekday vs Weekend Transaction Pattern

Across the row-level observations, average transactions are **1,252,082 on weekdays** versus **1,338,142 on weekends**.

KPI:

Highest transactions per merchant: **Rajasthan**

Transactions per merchant: **≈22.18**

Lowest transactions per merchant: **West Bengal**

Transactions per merchant: **≈17.71**

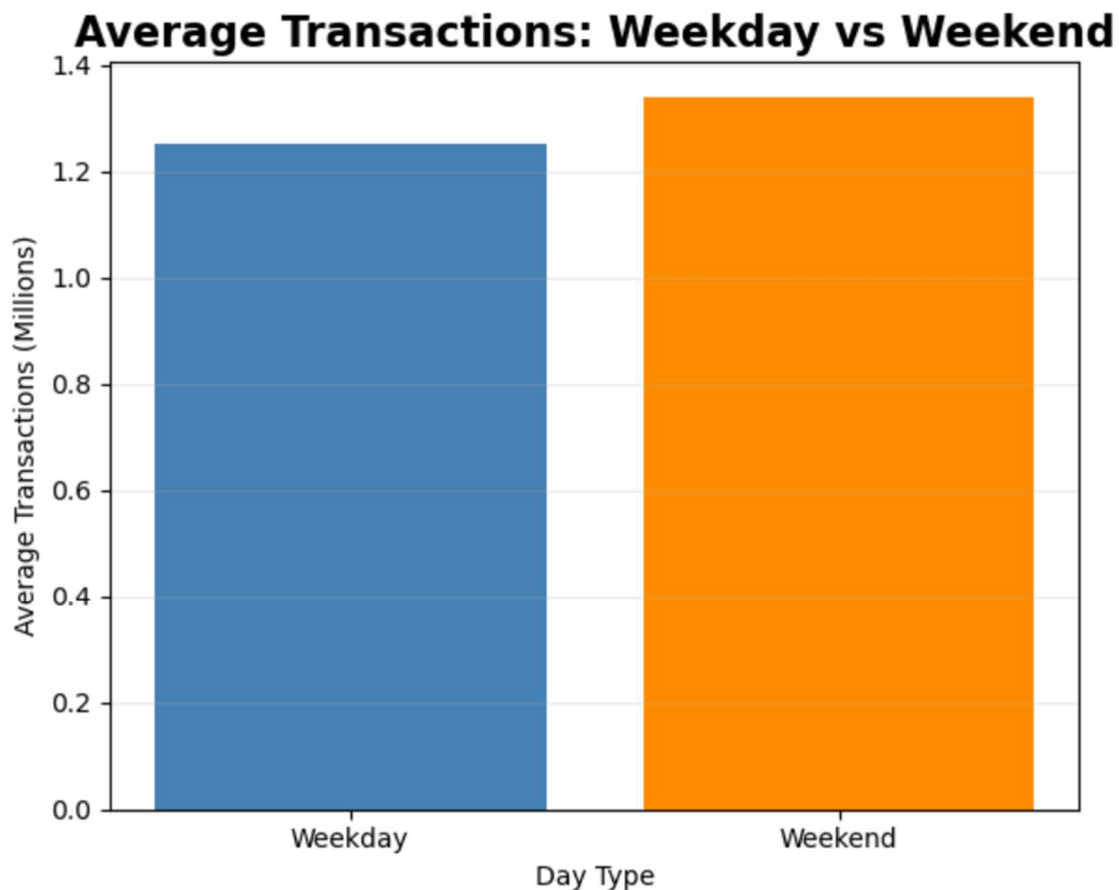


Figure 11. Weekdays vs Weekend

INSIGHTS:

- However, the dataset is not a complete continuous daily series for every state/category combination.
- Therefore, the result should be treated as a descriptive comparison of the available observations.

14. Customer Satisfaction (CSAT)

The overall row-level average **CSAT is 4.71/5**. Rajasthan has the **highest** average CSAT **at 4.78**,

West Bengal has the **lowest at 4.60**. CSAT is useful as a customer-experience KPI alongside operational measures such as success rate.

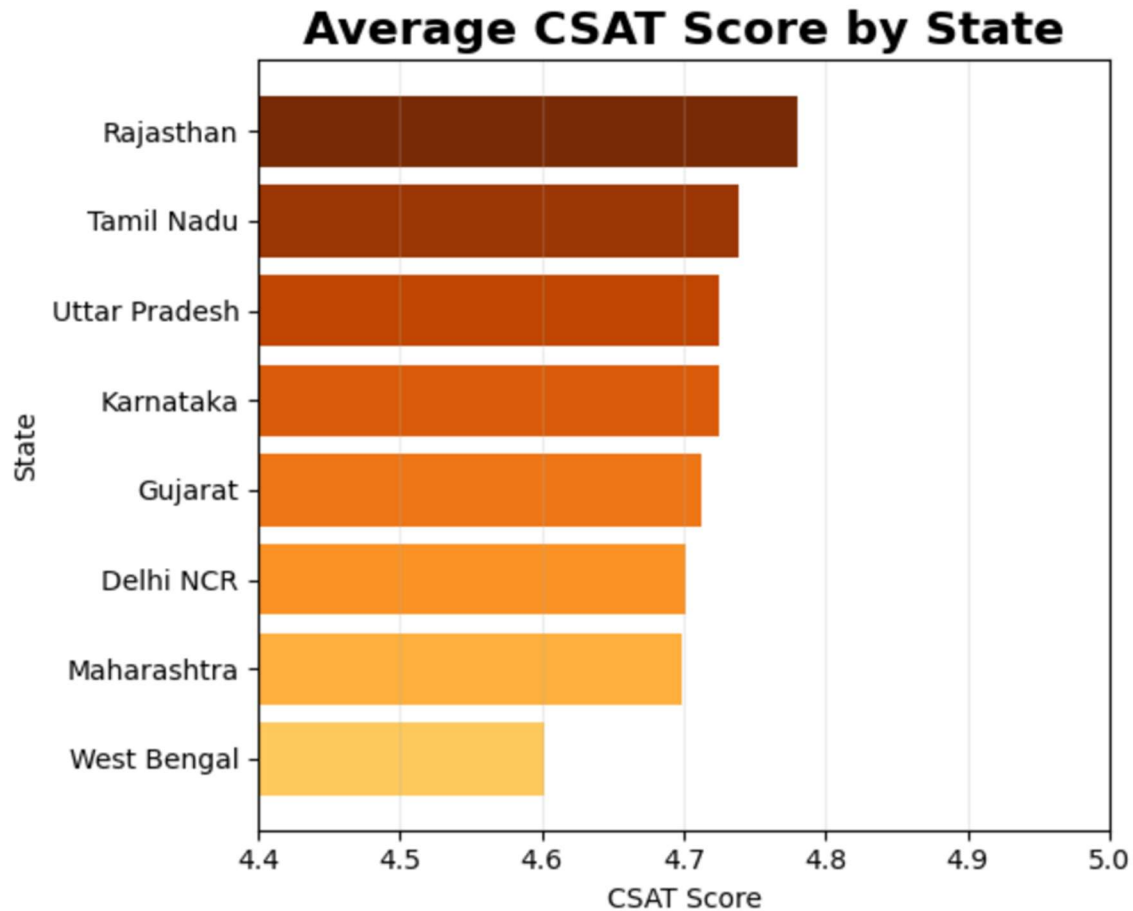


Figure 12. Average CSAT score by state

INSIGHTS:

- This shows some variation in customer satisfaction between states.
- Rajasthan represents the **strongest CSAT** performance among the states shown.
- West Bengal may require closer investigation to understand its comparatively **lower score**.
- CSAT should be considered alongside operational KPIs such as success rate.

KPI:

Highest CSAT state: **Rajasthan**

CSAT: **4.78 / 5**

Lowest CSAT state: **West Bengal**

CSAT: **4.60 / 5**

Summary of Key Metrics

Metric	Value
Total transactions	23,853,873
Total GMV	₹1,428.14 crore
Average success rate	97.96%
Average CSAT score	4.62 / 5
States tracked	8
Payment categories tracked	6
Dataset records	200
Downtime-flagged records	0

Next Steps to Move from Descriptive → Predictive

- **Demand forecasting** by state and payment category to estimate future transaction volumes.
- **Anomaly detection** for sudden changes in success rate, GMV or transaction volume.
- **Downtime-impact analysis** linking operational flags to transaction failures and lost GMV.
- **Merchant productivity** analysis using transactions per active merchant.
- **Trend and cohort analysis** when longer historical data becomes available.

This report answers what happened in the supplied dataset. The next analytical layer would use a longer time period and richer event-level data to explain why changes occurred and to forecast what is likely to happen next.

Glossary:

I. GMV — Gross Merchandise Value: Total monetary value of transactions processed/sold through a platform **before deducting fees, discounts, refunds, or other adjustments**. The exact definition can vary by dataset/platform.

II. UPI — Unified Payments Interface: India's **instant, real-time payment system** developed by NPCI that enables bank-account-to-bank-account payments.

III. ONDC — Open Network for Digital Commerce: An **open network/protocol for digital commerce** in India. Calling it simply a "government-backed open e-commerce protocol" is a little oversimplified. **Pin code can operate on the ONDC network**, but ONDC itself is not exactly a government-owned e-commerce platform.

IV. NPCI — National Payments Corporation of India: The organization that **operates** and **manages UPI** and **several other retail payment systems** in India.

V. CSAT — Customer Satisfaction Score: A customer-experience metric usually collected through **post-transaction or post-service surveys**, often asking customers to rate their satisfaction.